

Reinforcement Learning for Environment & Agent

Comprehensive Analysis of Dynamic Programming, Deep Q-Networks, Actor-Critic Architectures, and Model-Based Policy Optimization

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Course Code & Title **MLA0301 -**
Reinforcement Learning

Assessment Tool **CO4-AT1 -**
Technical Presentation

Executive Agenda & Module Overview

OVERVIEW



1. Dynamic Programming

Policy evaluation, policy iteration, value iteration, and Bellman's principle of optimality in sequential decision-making.



2. Deep Q-Networks

Deep Q-Network (DQN) architecture, stability mechanisms, experience replay memory, and target network decoupling.



3. Advanced DQN Variants

Double DQN, Dueling architecture, and Prioritized Experience Replay (PER) for sample efficiency and overestimation mitigation.



4. Actor-Critic & Model-Based

A2C/A3C continuous/discrete control, Model-Based RL frameworks, planning strategies, and industrial benchmarks.

TOPIC 01

Dynamic Programming for Sequential Decision-Making

Evaluating exact algorithms for prediction, policy evaluation, and policy improvement in known Markov Decision Processes (MDPs).

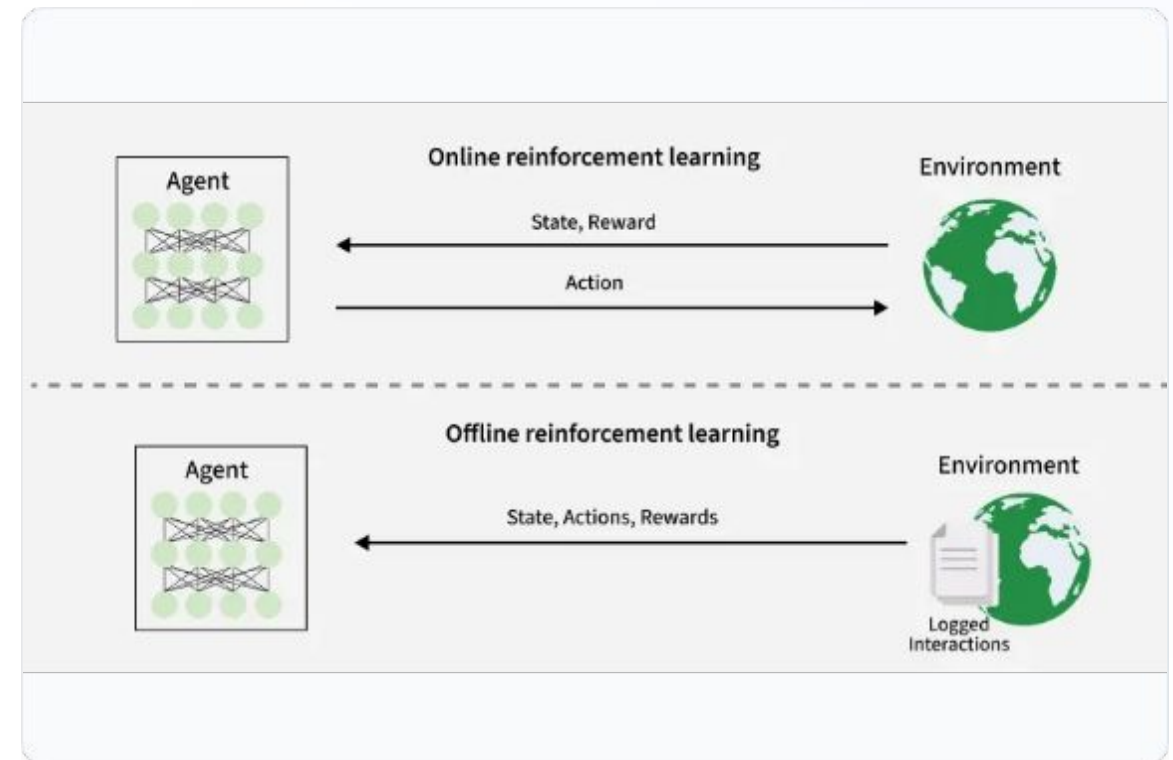
Bellman Optimality & DP Iteration Schemes

Q1 ANALYSIS

Bellman Optimality Equation

$$V^*(s) = \max_a \sum_{s', r} p(s', r | s, a) r + \gamma V^*(s')$$

- ✓ **Policy Iteration:** Alternates between full policy evaluation until convergence and greedy policy improvement.
- ✓ **Value Iteration:** Combines evaluation and improvement in a single step using the Bellman optimality backup operator.



Industrial Applications of Dynamic Programming

Q1 APPLICATIONS



Inventory & Supply Chain

Optimal stock replenishment policies under discrete stochastic demand profiles, minimizing holding and stockout costs over finite horizons.



Routing & Logistics

Shortest path finding, automated guided vehicle (AGV) routing in dynamic warehouses using modified Dijkstra and Bellman-Ford algorithms.



Energy Management

Optimizing battery storage charge/discharge cycles and micro-grid power dispatch based on deterministic forecasted load curves.

TOPIC 02

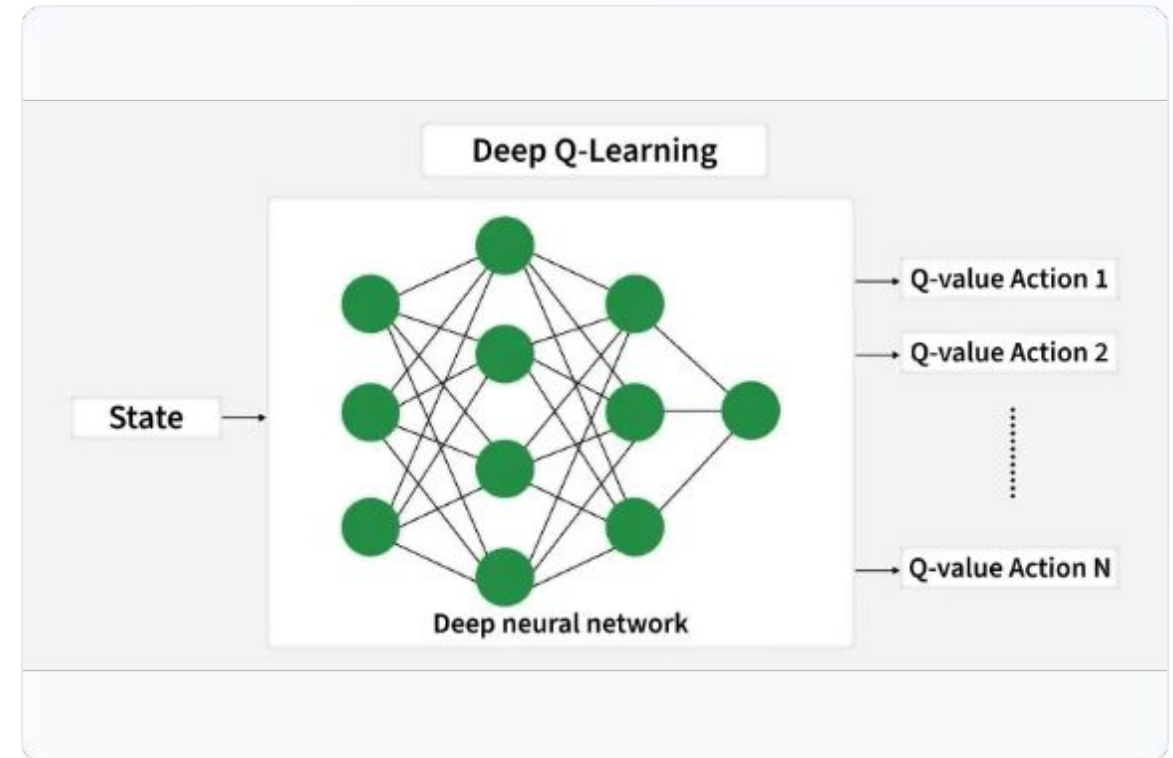
Deep Q-Networks (DQN) for High-Dimensional Decision-Making

Bridging traditional Q-learning with deep neural network function
approximators to solve continuous state-space problems.

DQN Architecture & Key Stability Mechanics

- Deep Neural Network:** Approximates action values directly from raw state inputs without hand-crafted features.
- Experience Replay:** Stores transitions in a buffer and samples mini-batches uniformly to break temporal correlation.
- Target Network:** Maintains frozen parameters updated periodically to prevent non-stationary target oscillation.

$$L(\theta) = \mathbb{E} \left[r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right]^2$$



TOPIC 03

Advanced Extensions of Deep Q-Learning

Overcoming overestimation bias, state-action decoupling, and uniform sampling inefficiency via DDQN, Dueling DQN, and PER.

DDQN, Dueling DQN & Prioritized Experience Replay

Q4 INNOVATIONS



Double DQN (DDQN)

Decouples action selection from value evaluation using Online Network for selection and Target Network for evaluation to eliminate systematic overestimation.



Dueling DQN

Splits network architecture into state-value stream and advantage stream, enabling faster learning of state values independent of actions.



Prioritized Replay (PER)

Samples transitions proportional to absolute TD-error, prioritizing informative updates while using Importance Sampling weights to correct bias.

Comparative Performance of Deep Q-Learning Algorithms

Q4 BENCHMARKS

Algorithm Variant	Key Modification	Primary Benefit	Sample Efficiency	Computational Overhead
Vanilla DQN	Experience Replay + Target Network	Stabilizes Deep Q-Learning	Baseline (Moderate)	Low
Double DQN	Decoupled Selection & Evaluation	Eliminates Value Overestimation	High (+25%)	Negligible
Dueling DQN	Dual Stream V(s) & A(s,a) Output	Better Generalization across actions	High (+35%)	Low (Slightly more parameters)
Prioritized PER	TD-Error Weighted Sampling	Focuses on critical/rare experiences	Very High (+50%)	Moderate (SumTree maintenance)

TOPIC 04

Actor-Critic Methods for Real-Time Systems

Combining policy-based variance reduction with value-based bias reduction
for scalable real-time control.

Advantage Actor-Critic (A2C / A3C) Framework

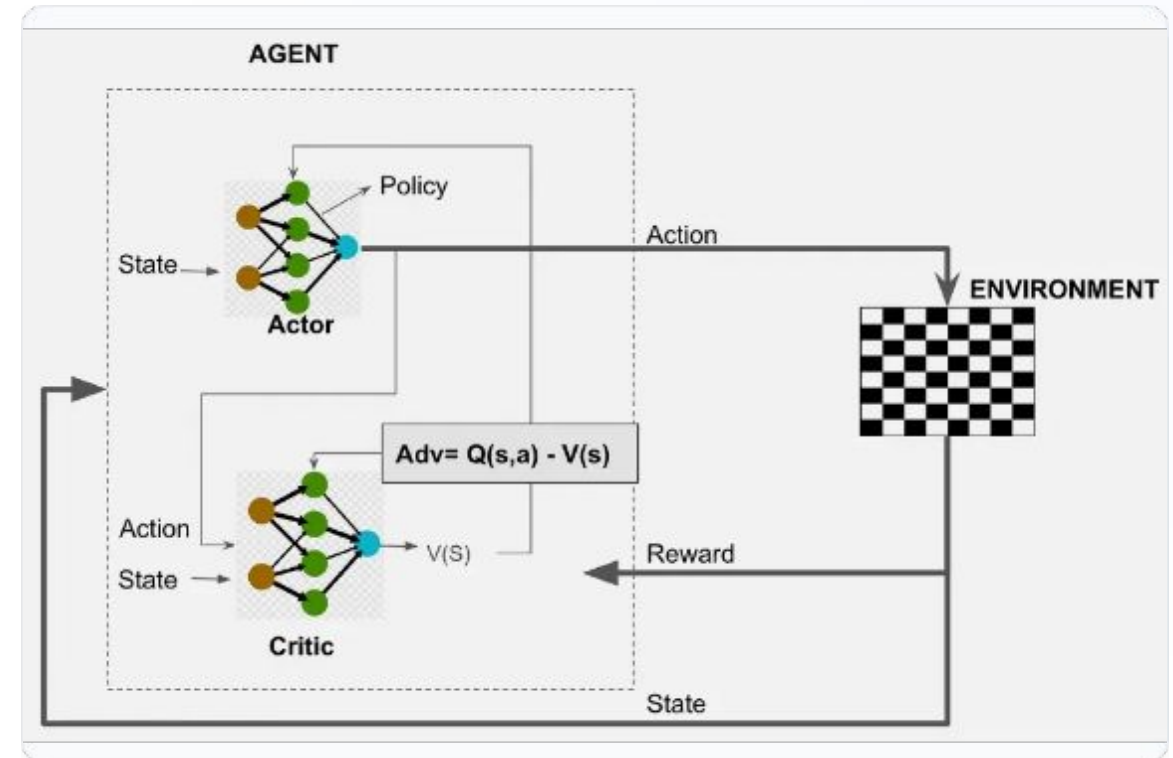
Q6 ACTOR-CRITIC

Actor (Policy Update): Updates policy parameters in direction of estimated advantage:

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[\nabla_{\theta} \log \pi_{\theta}(a|s) A(s, a; w) \right]$$

Critic (Value Estimation): Evaluates baseline state values using $V_w(s)$ TD-learning to compute advantage $A(s, a) = r + \gamma V(s') - V(s)$

A3C vs A2C: A3C uses asynchronous parallel worker threads without replay memory; A2C synchronizes gradient updates deterministically.



TOPIC 05

Model-Based Reinforcement Learning & Planning

Leveraging explicit transition kinetics and environment modeling for sample-efficient autonomous decision-making.

Model-Based RL Framework & Robotics

Q8 MODEL-BASED

- ⚙️ **Environment Dynamics Model:** Learns transition probability and reward
 $f(\text{model}) \rightarrow s'$ $R(s, a)$
- 🧩 **Decision-Time Planning:** Uses Model Predictive Control (MPC) and Monte Carlo Tree Search (MCTS) for dynamic rollout trajectories.
- 🤖 **Robotics Application:** Dramatically reduces sample complexity by generating simulated rollouts, enabling safe manipulation learning in physical robots.



Comparative Analysis of Core RL Frameworks

Q10 SYNTHESIS

Dimension	Value-Based (e.g., DQN)	Policy-Based (e.g., PPO)	Model-Based (e.g., MBPO)
Action Space	Discrete / Low-Dimensional	Continuous & High-Dimensional	Both Continuous & Discrete
Sample Efficiency	Moderate (Off-Policy Replay)	Low (On-Policy Samples)	Very High (Simulated Rollouts)
Convergence & Stability	Susceptible to divergence	High (Monotonic Bound)	Model Bias Dependent
Primary Industrial Domain	Game AI, Grid Optimization	Autonomous Driving, Drone Control	Industrial Robotics, Automation

TECHNICAL DEFENSE

Questions & Answers

Thank you for your attention during this technical evaluation presentation.

PRESENTER DETAILS

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Register No: 192472012 | Course: MLA0301 (Reinforcement Learning)

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Image Sources



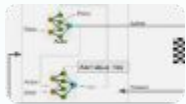
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